Soil: let the water through

Teacher guide and worked answers

Compare how much water passes through different soils in the same time.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsb/inputs/Build SOW 2026-2027.xlsx | BUILD Weekly - Spring | C31

Compare different soils and how easily water passes through them.

Direct match to C 31. Uses equal-input, fixed-time collected volumes as evidence of comparative drainage. Explicitly avoids claiming a laboratory permeability value or that one soil is universally best.

Prior learning

Soil contains mineral and organic material.

Pores are spaces that can contain air or water.

A larger number represents a larger volume when the units are the same.

Success evidence

Keep water input and collection time the same when comparing soils.

Read and compare collected volumes in mL.

Use results to name which sample let more water through in this test.

Equipment

Printed sample-data/evidence page, chosen response page and pencil; optional calculator.

Optional practical: three identical draining plastic pots, matched filters and stable supports, tray, 100 cm^3 scoop, 60 mL measure, timer and three graduated collection containers.

Prepared soil samples A, B and C from a known teaching source; one fresh portion per repeat.

Preparation

Pilot the method with the actual soils so pouring does not overflow. Select safe amounts and matching apparatus before class; if changing the model method, record the new input/time clearly.

Use approximately matched air-dry starting samples, prepared beforehand without creating dust, then loosely fill equal 100 cm^3 volumes without tamping. Equal volume does not imply equal mass.

Use identical filters prepared in the same way. If pre-wetted, allow them to drain for the same time and start each collection with an empty dry receiver.

Set up practical stations before the lesson. Allocate one sample per group and combine results, or use supplied constructed data.

Print one chosen response route. No pupil must finish every route.

Practical care

Use teacher-screened soils from a known teaching source; avoid contaminated soil, visible mould, sharp items and dust clouds.

Keep soil in trays, use spoons and wash hands afterwards. Do not taste or smell closely.

Use stable plastic pots and containers over a spill tray; wipe spilled water promptly.

Use room-temperature water. An adult prepares drainage holes and checks supports; pupils do not cut containers.

Ready to teach alternative

The constructed sample data supply two repeat results for A, B and C, using the complete printed method. Pupils can design the comparison, read volumes, judge fairness and draw a conclusion without handling soil.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	Which sample drains most?
2	Retrieval	2-5	Find the missing links
3	I do	5-8	Follow a connected path
4	I do	8-10	Give each soil the same test
5	I do	10-12	Read the scale carefully
6	We do	12-16	Repair the unfair test
7	We do	16-20	Read; compare; justify
8	Hinge	20-23	What can we conclude?
9	You do	23-25	Take a role; choose a route
10	You do	Within stage	Our method in six actions
11	You do	25-35	Make the evidence count
12	You do	Within stage	Check before you claim
13	Review	35-38	Lundy loop: improve our test
14	Exit	38-40	Leave with an evidence claim

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening Which sample drains most?

Present a familiar school-ground problem without assuming an actual site condition. Today we test samples, not decide the whole grounds design. Pupils can choose equipment or a paper lab.

2 Retrieval Find the missing links

Retrieve prior soil lesson but provide the answers and a brief recap if this is taught separately. Ask pupils to show the larger number privately.

Reveal: 1 Mineral and organic material. 2 A space between particles. 3 40 mL.

3 I do Follow a connected path

Trace the broad blue route. Think aloud: I am looking for a connected route, not counting how brown the soil looks. In real soils, roots, cracks, particle mixtures and compaction affect paths. Do not teach that clay has fewer total pores than sand; tiny pores can hold water while large connected pores let it flow more freely.

4 I do Give each soil the same test

Explain that the receiver sits directly below the draining pot in the actual setup; the diagram separates the parts for clarity. Demonstrate the method: start timer with first water, pour over 10 seconds, read at 2 minutes. Same pot/filter, loose packing and starting moisture. We compare water collected, a classroom indication of ease of drainage, not an intrinsic permeability coefficient.

5 I do Read the scale carefully

Think aloud: the scale goes 0, 20, 40, 60. The level is halfway between 20 and 40, so 30 mL. This is a flat-surface model. With clear water in a narrow cylinder, read the bottom of the curved meniscus at eye level; with cloudy drainage use the visible surface consistently. Choose apparatus with a readable scale.

Reveal: 30 mL: halfway between 20 and 40 mL.

6 We do Repair the unfair test

Work together as a test-design team. Each pupil selects one repair. B would confound water amount with soil; D would confound packing with soil. Same loose fill does not guarantee identical microscopic structure, but controls the action we take.

Reveal: Use A and C. Repair B: equal water. Repair D: the same loose preparation.

7 We do Read; compare; justify

Cover trial 2 briefly if helpful. Invite one learner to read, one compare, one justify. A> B> C in trial 1. Reveal trial 2 and note the same ordering. Constructed values illustrate reasoning; they are not real class results or universal values for named soil types.

Reveal: A: 42 mL. It is more than B: 26 and C: 10. Input and time were the same.

8 Hinge What can we conclude?

All pupils commit first. If they choose B/C, point to the phrase “in this test”. If data comparison is wrong, line up the values 42, 26, 10 on a number line. Proceed to an appropriately supported independent route.

A. A let most water through in this two-minute test. - A has the largest collected volume under the matched test conditions.

B. C is the best soil for every plant. - This test alone does not measure every plant need or all soil properties.

C. A must always drain fastest outdoors. - Outdoor structure, depth, cracks and starting moisture can change drainage.

9 You do Take a role; choose a route

Choice is about access and evidence. Practical groups can test one soil and share class results, while individuals complete their chosen page. Paper pupils use both supplied trials. All routes need a compared result and a conclusion.

10 You do Our method in six actions

Use identical draining pots and matching filters. Loosely fill each with 100 cm³ of prepared soil, without tamping. Add 60 mL over 10 seconds. Start timing when pouring starts. Collect and read at the stated time. Use fresh equally prepared portions for repeats. Keep soil out of the collection scale. Read at 2 minutes from the start of pouring. For practical pupils, a class trial plus the printed repeat can be compared but never presented as two class trials; label the source of each set.

11 You do Make the evidence count

Circulate during the working time. Use practical readings if valid and record spills/overflow as failed trials needing a repeat; do not replace inconvenient data silently. Pupils using paper data must label them constructed sample data.

12 You do Check before you claim

Shares the independent time. Prompt a self-correction rather than adding another worksheet. A conclusion can be spoken and adult-scribed.

13 Review Lundy loop: improve our test

Act on one pupil suggestion by adding it to the class method. State the influence: “We are changing our instruction because you spotted…” If using supplied data only, revise the model protocol, not pretend a real trial occurred.

Reveal: For example: matched preparation, an equal pour time, accurate readings or fresh repeated trials.

14 Exit Leave with an evidence claim

Record core success when pupil compares a volume and identifies a matched condition. If not secure, reteach with just A and C and concrete number cards before the retention lesson.

Reveal: A. For example, 60 mL input or 2 minutes. No: more evidence about plant needs is needed.

Answers and assessment

Retrieval 1-3

Mineral and organic material; a space between particles; 40 mL is greater than 20 mL.

I do scale

30 mL, halfway between 20 and 40. Use unit. The drawn scale is simplified; model labels are not a calibration for a real container.

W1 planning cards

Use A and C. Repair B by giving all samples 60 mL. Repair D with the same loose fill/no tamping for each. Also match pot/filter, soil volume, starting moisture and pouring duration.

We do data / hinge

Trial 1: A 42>B 26>C 10 mL. Trial 2: A 40>B 28>C 12 mL. A let most through under these conditions. The correct hinge is A in this two-minute test. Claims about best soil for all plants or always-fastest outdoor drainage overreach the evidence.

S1-S5

S1 42 mL. S2 A because 42 mL is greater than 26 and 10. S3 equal input and time. S4 check whether a similar pattern occurs/reduce trust in an accidental error; repeated results need not be identical. S5 water passes through connected pores between particles.

T1-T2

T1 state constructed data or actual class measurement; keep input 60 mL, time 2 min, soil volume 100 cm^3, pot/filter, pour 10 s and preparation matched. T2 for supplied trial 1: all input 60, time 2; outputs A 42, B 26, C 10 mL. Actual readings can differ; do not force them to match sample data.

T3-T5

T3 supplied: A 42>B 26>C 10 mL so A drained most in this test. For actual results use the recorded values and qualified conclusion. T4 yes, A>B>C in both constructed trials. T5 accept spillage, blocked holes, mismatched packing, wrong time, poor volume reading, or improving controls/repeating fresh samples. A spoiled trial is flagged and repeated; it is not quietly rewritten.

X1-X4

X1 both trials have A>B>C; A 40-42, B 26-28, C 10-12 mL. X2 packing changed; the tightly packed sample collected less, consistent with restricted pathways in this model test. X3 either change could cause the result, so their effects cannot be separated. X4 water held, nutrients, pH, root growth, plant type or relevant field conditions; do not claim the fastest-draining soil is automatically best.

E1-E3

E1 A with 42 mL trial 1 or 40 mL trial 2. E2 any valid matched condition. E3 no; drainage alone does not cover water storage, plant needs, nutrients and other conditions.

R1-R2

Accept a relevant improvement linked to evidence or measurement. Role preference has no fixed correct answer; value a practical reason or comfort/access need.

Optional practical

Actual outcomes vary. Repeat with fresh portions if feasible; identify the source of every comparison. Equal input and equal time are core controls, with sample volume, packing, starting moisture, filter and container controls needed for a credible comparison.

Teaching and assessment notes

40 minutes: opening 2, retrieval 3, I do 7, We do 8, hinge 3, You do 12, review 3, exit 2. Zero-minute slides share the independent stage.

The word permeability is introduced qualitatively. This falling-head, partly unsaturated classroom pot test reports collected volume at a fixed time; it is not a standard laboratory measurement of permeability or field infiltration rate.

The actual receiver sits under the pot. The editable diagram separates it to keep labels clear. Always use a stable support and a spill tray.

Teacher script: “I can compare 42, 26 and 10 because each trial started with the same input and stopped at the same time. I will say in this test, because the real ground can behave differently.”

Avoid rigid sand-versus-clay rules. Connected large pores, cracks, roots, compaction, starting water content and particle mixtures influence results. The supplied A/B/C are unnamed teaching samples so pupils must use evidence rather than a memorised stereotype.

For learners with a fragile grasp of number magnitude, use only A 42 and C 10 first; then return to B 26. Preserve the same science objective.

If practical measurement fails, label it honestly and use the supplied constructed data for the reasoning task. Doing the calculation from a paper dataset can demonstrate the same comparison skill.

Access and responsive teaching

Supported

Use the two-minute data and number comparison. Adult reads units; pupil matches largest/smallest and gives an evidence sentence or points to values.

Standard

Record two results, compare the pattern and name controlled conditions. Optional practical role: timer or reader.

Stretch

Evaluate a confounded test, compare repeat patterns and explain why field drainage may differ.

Regulation

Choose a dry role or paper lab; show the predictable two-minute window. No race, compulsory handling or public speed scoring. Pair pupils by chosen role, not fixed ability label.

Misconceptions to check

The tallest water level always means the most water.

Only compare heights directly in identical containers; read the volume scale in mL.

Check: Ask why a narrow and wide cup could have different heights for the same volume.

We can give different amounts of water because the soils are different.

Keep water input equal to compare the soil effect.

Check: Ask pupils to repair a 60 mL versus 100 mL test.

Large soil particles have no spaces between them.

Water moves through connected spaces; the size and connection of spaces matter.

Check: Trace a water path through the model.

The sample with least drainage must be the best soil.

The best soil depends on the job. Drainage alone cannot tell us everything about plant growth.

Check: Ask what else a grower would need to know.

Word	Meaning
drain	For water to move through and out.
permeable	Allowing water to pass through.
volume	The amount of space taken up; here measured in mL.
fair comparison	Changing the factor being tested while keeping other important conditions the same.
repeat	Do the test again to check whether the pattern is similar.

Scientific references

USDA NRCS: Soil Infiltration — Guides for Educators

Checked 8 September 2026. Supports effects of pores, texture, initial moisture and compaction on water movement. Our pot drainage test is a comparative classroom model, not a measurement of field infiltration or intrinsic permeability.

USDA NRCS: Soil Bulk Density, Moisture and Aeration — Guides for Educators

Checked 8 September 2026. Supports pore space, stored water, organic matter and the distinction between water held and water available to plants. Short-time classroom water balance is an estimate and not a field-capacity measurement.